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Chapter 16
Small Bets, Big Pots: No-Fold [0,1] Games

In Chapter 15, we solved our first full-street game, where both players have the option to bet. There, we returned to consider the AKQ game. In this chapter, we return to a game that we considered when we began looking at half-street games; the [0,1] game. The [0,1] game is among the most important games we consider, because it is the game that is most closely related to poker, primarily because of the many hand types and the necessity of utilizing betting regions.

Throughout this chapter, all of the games we consider will have a special rule that we first introduced in Chapter 11, which is that neither player may fold. The "neither player may fold" rule approximates play when the pot is extremely large. What this means for us here is that none of the strategies that follow contain bluffing; such strategies would be of no use, since there is no opportunity for folding.

No-fold games are exercises in extracting the proper amount of value from strong hands.

The first and simplest [0,1] full street game is where there is only one bet remaining (i.e., no raising is allowed). Hence, there is only one decision for each player. X must decide what hands to bet, and Y must decide what hands to bet if X checks. If either player bets, the other player must call.

Before going on to the solution, see if you can figure out the optimal strategies; this is a good method of improving intuition.

Example 16.1 - [0,1] Game #4

One full street.
Neither player may fold.
Only one bet is allowed.

In this game, there is only one bet available, so the number of possible betting sequences is three:

X bets-Y calls.
X checks - Y bets - X calls.
X checks - Y checks.

Solving the problem intuitively, X is going to bet at least some of the time. If he bets, he will never bet a hand that is worse than a hand he would check, since Y can never fold. If X checks, however, Y can bet all the hands that X would have bet, since he is sure to win (given that X checked). In addition, Y can bet the strongest half of the remaining hands. If X's hand is in that same region, then Y will break even on X's calls (overall), but if X's hand is the lower half of that region, then Y will win a bet.

To recap Y's strategy:

If X bets, then Y's action is forced (he must call).
If X checks, then Y responds by betting all hands that X would have bet, and the best half of the hands that he would check.

Given this strategy for Y, then, what is X's strategy?

Y gains value when he bets a hand after X checks. Otherwise, Y's strategy is forced. So to

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prevent Y from exploiting him, X can simply bet all hands. If he does this, he guarantees that the game will have equal equity for both players (one bet will go in).

So the optimal strategy for this game is for X to bet every hand, and Y to call every hand.

Analytically, recall that we referred to thresholds earlier as the key components of the solutions to our strategies. In fact, even in this simple game we solve for the thresholds of each strategy.

We parameterize the optimal strategies for both sides as follows:

There is a threshold x_1 such that below this threshold, X will bet, and above this threshold, X will check.

There is a threshold y_1 such that below this threshold, Y will bet if checked to, and above this threshold, Y will check.

To make X's strategy optimal, X must make it so that Y cannot unilaterally improve his equity by changing y_1, and vice versa. Each player, then, wants to play so that the other is indifferent between the two actions on either side of each of his threshold points. This indifference means that the equity of the two actions is equal. We showed earlier how we can solve games by application of this concept. We write indifference equations for each threshold point.

Recall that we are referring to ex-showdown expectation here; only the change in equity that occurs as a result of the postflop betting is considered.

The indifference at y_1 (where Y is indifferent between checking and betting):

X's hand	***p*(X's hand)**	**<Y, bet>**	**Product**	**<Y, check>**	**Product**
$[0, x_1]$	x_1	N/A	N/A	N/A	N/A
$[x_1, y_1]$	$y_1 - x_1$	-1	$(x_1 - y_1)$	0	0
$[y_1, 1]$	$1 - y_1$	+1	$(1 - y_1)$	0	0
Total			$\mathbf{x_1 - 2y_1 + 1}$		0

As we did in Example 11.3 in Chapter 11, we sum the total of the "Product" columns, and set the weighted expectations equal:

$0 = x_1 - 2y_1 + 1$
$2y_1 = x_1 + 1$

We next consider the indifference at x_1.

At x_1 (where X is indifferent between checking and betting):

X's hand	***p*(X's hand)**	**<Y, bet>**	**Product**	**<Y, check>**	**Product**
$[0, x_1]$	x_1	-1	x_1	-1	x_1
$[x_1, y_1]$	$y_1 - x_1$	+1	$(y_1 - x_1)$	+1	$y_1 - x_1$
$[y_1, 1]$	$1 - y_1$	+1	$(1 - y_1)$	0	0
Total			$\mathbf{1 - 2x_1}$		$y_1 - 2x_1$

Setting the expected values equal and solving for y_1.
$1 - 2x_1 = y_1 - 2x_1$
$y_1 = 1$

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Substitution:

$2\ y_1 = x_1 + 1$
$2 = x_1 + 1$
$x_1 = 1$

This is the same answer we found intuitively; both x_1 and y_1 are located at 1. So X bets all hands, and Y never gets a chance to bet; he must simply call.

Example 16.2 - [0,1] Game #5

The next simplest [0,1] full-street game introduces the concept of raising for value and has the following parameters:

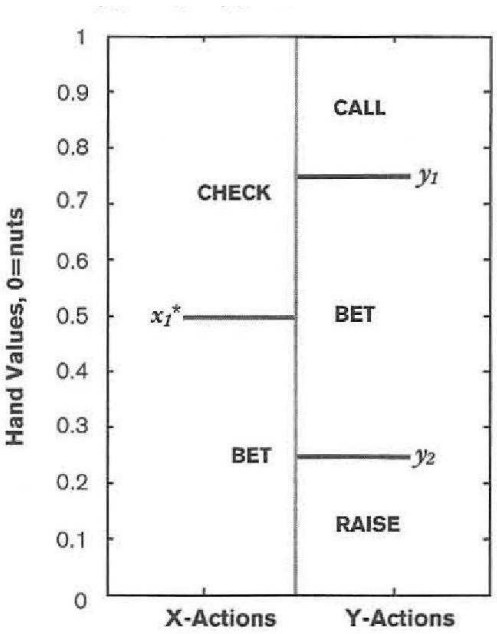
One full street.
Two bets left.
Check-raising is not allowed.
Neither player may fold.

In analyzing this game we will use a notation that we have alluded to and which should seem intuitive; the threshold between betting and raising, y_2 There is no x_2 m this game, as due to the impermissibility of check-raise, X can never put in the second bet. However, when X bets, Y will raise with some fraction of the hands X bets.

In Game #4 (Example 16.1), we saw that X can prevent Y from exploiting him by simply betting all his hands. However, this strategy will not do here, as Y has the option of raising and X can improve on his equity with his very worst hands by checking and calling instead of betting.

The first step is to parameterize the solution:

$0 < y_2 < x_1 < y_1 < 1$



16.1 Strategy structure for [0,1] Game #5

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What this parameterization means is that the optimal strategies will have the following form: X will bet some amount of hands. Y will raise with fewer hands than that. If X checks, Y will bet more hands than X would have.

We can then write indifference equations at each threshold value:
In our table, we can simplify the process by identifying only the difference between one EV and the other for each row. Then we can make all the differences add up to zero.

At y_2 (the indifference between Y calling and Y raising):

X's hand	**Y calls**	**Y raises**	**Y difference**	**Product**
$[0, y_2]$	-1	-2	1	y_2
$[y_2, x_1]$	+1	+2	-1	$-1(x_1 - y_2)$
Total				$\boldsymbol{2y_2 - x_1}$

Recall again that Y's equity from calling or raising must be the same due to our principle of indifference. Likewise, the lengths of the intervals $[0, y_2]$ and $[y_2, x_1]$ represent probabilities because the hands were originally' drawn from a uniform distribution.

$2y_2 - x_1 = 0$
$y_2 = x_1 / 2$

This indicates that Y will raise with half the hands that X bets. The reasoning for this is similar to the reasoning that we saw in the half-street game. In an no-fold game, if X holds some number of hands, Y can put in one additional bet. X must call and may not fold. Whether the bet Y puts in is the first or second bet, Y puts it in with half the hands that X can hold.

Even in cases where X can reraise, however, we find that in these no-fold games that there is a ratio between the hands that X can hold and the hands that Y will put in the next bet. We will use the variable *R* to represent the ratio between the two:

$y2 = Rx_1$

This may seem like an unnecessary complication, but it will be prove useful. R is an important concept, representing the fraction of X's value betting hands that Y should raise. Here *R* is $^1/_2$, because X cannot reraise. In fact, this is always the case on the last bet - when a player cannot be reraised and the other player cannot fold, *R* is $^1/_2$.

At x_1, we see the effect that Y's ability to raise has on X's betting thresholds. As betting exposes X to being raised, he is forced to bet a narrower distribution of hands.

For X at x_1 (the indifference between X checking and betting):

Y's hand	**X check-calls**	**X bets**	**Difference**	**Product**
$[0, y_2]$	-1	-2	+1	y_2
$[y_2, x_1]$	-1	-1	0	0
$[x_1, y_1]$	-1	-1	0	0
$[y_1, 1]$	0	+1	-1	$-(1 - y_1)$
Total				$\boldsymbol{y_2 - 1 + y_1}$

Writing the indifference equation here results in an equation that lacks any x values at all; what it

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states is that the size of the raising region of Y's optimal strategy must be equal to the size of the region he checks behind when X checks.

$y_2 = 1 - y_1$

At y_1, we see the familiar pattern of Y betting behind X's check with $^1/_2$ of the hands that X would check there:

At y_1 (the indifference between Y checking and Y betting):

X's hand	**Y check**	**Y bets**	**Y difference**	**Product**
$[0, x_1]$	N/A	N/A	N/A	N/A
$[x_1, y_1]$	0	-1	+1	$y_1 - x_1$
$[y_1, 1]$	0	+1	-1	$-1(1 - y_1)$
Total				$\boldsymbol{2y_2 - x_1 - 1}$

$2y_2 - x_1 - 1 = 0$
$y_1 = (1 + x_1)/2$

We now have three equations and three thresholds:

$y_2 = Rx_1$
$y_2 = 1 - y_1$
$y_1 = (1+x_1)/2$

To solve, we set the first two expressions for y_2 equal and substitute the third expression for y_1:

$Rx_1 = 1 - (1+x_1)/2$

$$x_1 = \frac{1}{1 + 2R}$$

We know from earlier that $R = {}^1/_2$ Substituting, we have:

Solution:

$x_1 = {}^1/_2$
$y_2 = {}^1/_4$
$y_1 = {}^3/_4$

To verify that this solution is indeed optimal, we can examine the play and see if either player can improve by playing differently against the opponent's optimal strategy.

Consider X's play when this hand is x between 0 and $^1/_4$. His choices are to check and call or to bet. If he bets, then the following outcomes occur:

When Y is better than x, X loses two bets. $(p = x)$

When Y is worse than x but better than $^1/_4$, X gains two bets. $(p = (^1/_4 - x))$

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When Y is worse than $^1/_4$ X gains one bet. ($p = {}^3/_4$)

<y, x bets> = (x)(2) + ($^1/_4$ - x)(-2) + ($^3/_4$)(-l)
<y, x bets> = 4x - $^5/_4$

If X checks a hand that is between 0 and $^1/_4$.

When Y is better than x, X loses one bet. (p = x)
When Y is worse than x but better than $^1/_4$, X gains one bet. ($p = {}^1/_4 - x$)
When Y is worse than $^1/_4$ but better than $^3/_4$, X gains one bet. ($p = {}^1/_2$)
When Y is worse than $^3/_4$, X gains 0. ($p = {}^1/_4$)

<x, check> = (x)(l) + ($^1/_4$ - x)(-l) + ($^1/_2$)(-l)
<x, check > = 2x - $^3/_4$

This second value is greater than the first (Y's equity is greater) when:

2x - $^3/_4$ > 4x - $^5/_4$
x < $^1/_4$

So we can see that for X's hands between 0 and $^1/_4$, betting yields higher equity than checking - hence, X cannot improve by switching strategies. We could go on and check all the regions, looking for hands where we can unilaterally improve one player's equity. However, since the strategies we found here are optimal, no such opportunities exist.

In this last section, we also touched on an important concept - ***the game value***. Playing this game has value to Y, because of his positional advantage. In Chapter 18 we will look at game values in depth, especially comparing game values from different types of games to each other.

Example 16.3 - [0,1] Game #6 - The Raising Game

One full street.
Players are allowed to raise and reraise without limit and have arbitrarily large stacks.
Check-raise is not allowed.
Neither player may fold.

This game is similar to the previous game, except that instead of just two bets, we now allow the number of bets to grow forever. Of course, in our optimal strategies, one or the other player will eventually stop raising and call.

Our thresholds can be described as follows:

We have x_1, which is the threshold where X is indifferent to checking and betting. Since check-raise is not allowed, when X checks, there is just one remaining threshold y_1. We know from our discussion in the previous game that y_1 will be half the distance from x_1 to 1, because this is the last bet and X cannot raise. We also know from the previous discussion that X will never "bluff" by putting in the nth bet with a hand weaker than a hand that would not put in that bet.

When X does bet, however, we have an infinite series of thresholds, beginning with y_2, the _ threshold between hands that will raise and hands that will simply call. To the left of y_2, we have a threshold where X reraises x_3, and to the left of that, a threshold where Y puts in the fourth bet,

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y_4, and so on.

So our parameterization looks like this:

$0 < y_{2n} < x_{2n-1} < \ldots < y_2 < x_1 < y_1 < 1$

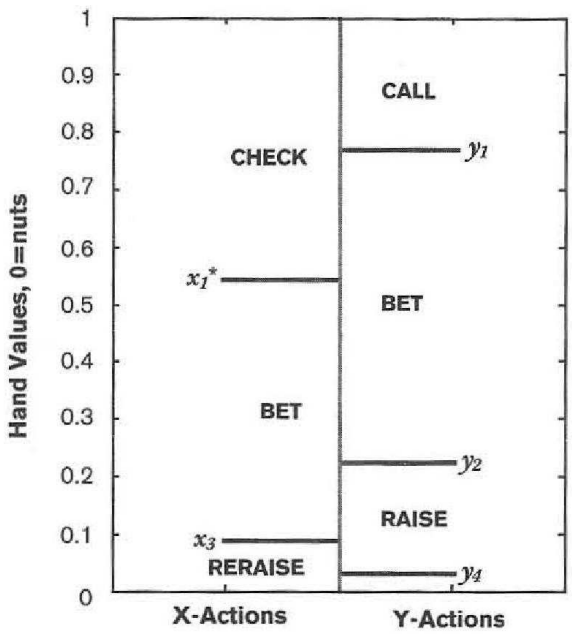

Figure 16.2 Strategy structure for [0.1] Game #6

This game has a sort of symmetry to it. First, consider the case where X bets. Y will raise with some fraction of his hands. Call this fraction (as before) R.

$y_2 = Rx_1$,
(just as in the previous game)

However, in this game, R is not $^1/_2$; X can reraise. Consider the situation once Y has raised. Y holds a hand in the range $[0, Rx_1]$. Now X wants to put in a raise with a fraction of the hands Y raised with. The situation for X here is the same as the situation for Y was on the previous bet - Y has a range, X wants to raise; his raising standard will be the same as Y's - he will raise with R of the hands that Y has.

$x_3 = Ry_2$

And now Y faces a bet from X, and will raise with R of the hands X has, and so on. Generally, then, for $n > 1$:

$y_n = Rx_{n-1}$ (note that y_3, y_5, etc., do not exist)
$x_n = Ry_{n-1}$ (note that x_2, x_4, etc., do not exist)

We have called [0,1] Game #6 "the raising game," because it contains the answer to a frequently asked question about poker - "how often should I raise my opponent if he can reraise me and no one can fold?" If we solve for R, we will know the proper frequency and be able to answer this question.

Next, consider the situation where X checks. We know that Y will bet with all the hands better

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than x_1 and half the hands worse than x_1:

$y_1 = (1+x_1) / 2$

Additionally, the same indifference at y_1 holds from the previous game (as there is no check-raising), leading to the size of Y's raising region y_2 being the same size as his checking region:

$y_2 = 1 - y_1$

Substituting:

$y2 = l - (l + x_1)/2$
$2Rx_1 = 2 - 1 - x_1$
$x_1 = 1/(2R + 1)$

Using this value, we can now find the values of all the thresholds y_2, x_3, y_4 and so on, in terms of R by multiplying by R the appropriate number of times. Additionally, we can find y_1 in terms of R.

But what is the value of *R?* For this we need one more indifference equation:

For Y at y_2 (indifference between raising and calling):

X's hand	**Y raises**	**Y calls**	**Y difference**	**Product**
$[0, x_3]$	-3	-1	+2	$2x_3$
$[x_3, y_2]$	-2	-1	+1	$y_2 - x_3$
$[y_2, x_1]$	+2	+1	-1	$- 1(x_1 - y_1)$
Total				$\boldsymbol{x_3+2y_2 - x_1}$

$x_3+2y_2 - x_1 = 0$

This is an indifference equation that relates three thresholds. However, we have all these things in terms of R and x_1 (for convenience):

$R^2x_1 + 2Rx_1 - x_1=0$
$R^2 + 2R - 1 = 0$
$R = \sqrt{2} - 1$

Recall that we christened $\sqrt{2} - 1$ as "the golden mean of poker." (See Equation 14.1.) Here it appears again as the answer to our question: "How often should I raise an opponent who cannot fold and can raise me back?"

So the solution to our game is as follows:

Solution:

$x_1 = l/(l+2r)$	≈ 0.5469
$y_1 = (1/(1 + 2r) + l)/2 = (1 + r)/(l + 2r)$	≈ 0.7735
$y_2 = r/(l+2r)$	≈ 0.2265
$x_3 = r^2/(1 +2r)$	≈ 0.0938

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$y_n = r^{n-1}/(1 + 2r)$ (for even $n > 1$)
$x_n = r^{n-1}/(1 + 2r)$ (for odd $n > 2$)

In our next game, we give Player X yet another option - check-raise.

Example 16.4 - [0,1] Game #7

One full street.
Two bets left.
Check-raising is allowed.
Folding is not allowed.

In Game #5 (Example 16.2), the two-bet case without check-raising, recall that there was no threshold x_2. When check-raise was not allowed, x_2 was meaningless, as there was no sequence of betting that would lead to X putting in the second bet. However, when we give X the option to check-raise, we have the same game, but with one additional threshold for which we must solve.

The key to this game is to identify hands with which X wishes to check-raise. Clearly, X wants to put in the second bet with hands that are better than y_1 - obviously hands that are worse than this would lose value by check-raising. Clearly, X will not put the second bet in with a hand weaker than he would put in the first - we've shown that this is dominated already. Intuitively, it would seem that X would want to be the one to put in two bets with his strongest hands; although, it turns out that X's strategies are co-optimal as long as he chooses the proper region size for check-raising with hands better than y_2. However, for simplicity, and because it aids us when more bets are available, we parameterize the game as follows:

$0 < x_2 < y_2 < x_1 < y_1 < 1$

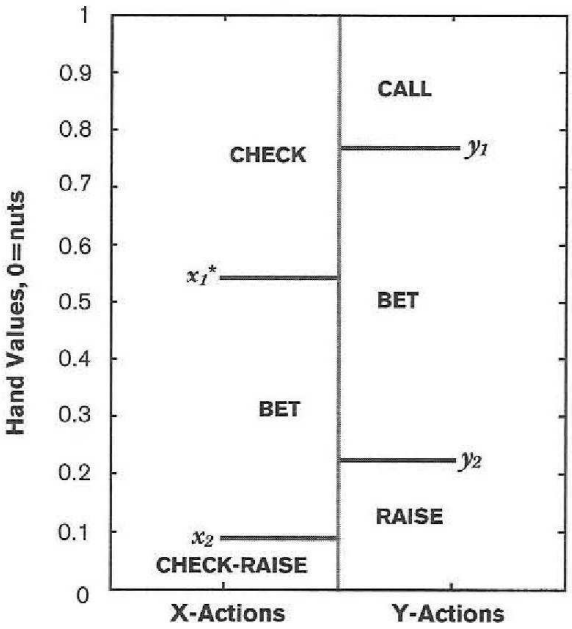

Figure 16.3 Strategy structure for [0,1} Game # 7

This translates to the following strategies.
For X:

X checks and raises if Y bets on $[0, x_2]$.

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X bets for value on $[x_2, x_1]$.

X checks and calls on $[x_1, 1]$.

For Y:

If X bets:

Y raises for value on $[0,y_2]$.
Y calls on $[y_2, 1]$.

If X checks:
Y bets for value on $[0, y_1]$.
Y checks on $[y_1, 1]$.

Creating indifference equations for the four thresholds:

For X at x_2 (indifference between check-raising and betting):

Y's hand	**X check-raises**	**X bets**	**Difference**	**Product**
$[0, y_2]$	-2	-2	0	0
$[y_2, y_1]$	+2	+1	+1	$y_1 - y_2$
$[y_1, 1]$	0	+1	-1	$-(1 - y_1)$
Total				$\boldsymbol{2y_1 - y_2 - 1}$

$2y_1 - y_2 - 1 = 0$
$y_2 = 2y_1 - 1$

For Y at y_2 (indifference between raising and calling):

X's hand	**Y raises**	**Y calls**	**Difference**	**Product**
$[0, x_2]$	N/A	N/A	N/A	N/A
$[x_2, y_2]$	-2	-1	+1	$y_2 - x_2$
$[y_2, x_1]$	+2	+1	-1	$-1(x_1 - y_2)$
$[x_1, 1]$	N/A	N/A	N/A	N/A
Total				$\boldsymbol{2y_2 - x_1 - x_2}$

$2y_2 - x_1 - x_2 = 0$
$y_2 = (x_1 + x_2)/2$

For X at x_1 (indifference between check-calling and betting):

Y's hand	**X check-calls**	**X bets**	**Difference**	**Product**
$[0, y_2]$	-1	-2	+1	y_2
$[y_2, y_1]$	-1	-1	0	0
$[y_1, 1]$	0	+1	-1	$-(1 - y_1)$
Total				$\boldsymbol{y_2 - 1 + y_1}$

$y_2 - 1 + y_1 = 0$
$y_2 = 1 - y_1$

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For Y at y_1 (indifference between betting and checking):

X's hand	Y raises	Y calls	Difference	Product
$[0, x_2]$	0	-2	+2	$+2x_2$
$[x_2, x_1]$	N/A	N/A	N/A	N/A
$[x_1, y_1]$	0	-1	-1	$-1(y_1 - x_1)$
$[y_1, 1]$	0	+1	+1	$1 - y_1$
Total				$\boldsymbol{2x_2 + 2y_2 - x_1 + 1}$

$2x_2 + 2y_2 - x_1 + 1 = 0$
$2y_1 = 1 + x_1 - 2x_2$

We now have four indifference equations, and we can solve for the four unknowns:

$2y_1 = 1 + x_1 - 2x_2$
$y_2 = (x_1 + x_2)/2$
$y_2 = 2y_1 - 1$

$y_2 = 1 - y_1$

$2y_1 - 1 = 1 - y_1$

$y_1 = {}^2/_3$
$y_2 = {}^1/_3$
${}^4/_3 = 1 + x_1 - 2x_2$
${}^1/_3 = (x_1 + x_2)/2$
$x_1 = {}^2/_3 - x_2$
${}^4/_3 = 1 + {}^2/_3 - 3x_2$
$3x_2 = {}^1/_3$
$x_2 = {}^1/_9$
$x_1 = {}^5/_9$

Solution:

$x_1 = {}^5/_9$
$y_1 = {}^2/_3$
$x_2 = {}^1/_9$
$y_2 = {}^1/_3$

We can compare this game to Game #5 (Example 16.2), the similar game without check-raise. What would we expect to be different? First, we would expect Y to bet fewer hands when X checks. This is because by betting in Game #5, Y was only exposing one bet. However, in Game #7, Y exposes a second bet, because sometimes X will check-raise. So Y must bet fewer hands behind X's check.

But this, in turn, causes X a problem. In Game #5, X was checking and calling with hands higher than but near ${}^1/_2$. These hands were gaining check-call value because Y was betting hands as weak as ${}^3/_4$. In the new game, however, these hands of X's lose some of that value because Y

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will no longer bet those mediocre hands for value. So X will likely bet additional hands worse than $^1/_2$. This in turn will allow Y to loosen up his raising threshold, since two things have occurred. First, X is checking some of his best hands, so Y does not have to fear X having those hands. Secondly, X is betting additional mediocre hands near $^1/_2$. Y can therefore raise for value with somewhat weaker hands in this game than in Game #5.

Inspecting the thresholds, we see that these changes are in fact reflected in the solution to the game:

Threshold	**Game #5**	**Game #7**
x_1	$^1/_2$	$^5/_9$
y_1	$^3/_4$	$^2/_3$
y_2	$^1/_4$	$^1/_3$

Our last full-street no-fold game is the analogue to Game #6 with check-raise: this game might be thought of as the full no-fold one-street [0,1] game.

Example 16.5 - [0,1] Game #8: The Raising Game with Check-Raise

One full street.
Players are allowed to raise and reraise without limit and have arbitrarily large stacks.
Check-raise is allowed.
Neither player may fold.

The final no-fold game in the one-street cases is the game where both players can put in as many raises as they think their hands merit and check-raise is allowed. Like Game #6 (Example 16.3), in this game we have an infinite number of thresholds, and so we cannot simply write all the indifference equations for this game. However, we can create a recursion for higher-valued indifference equations and solve that to find the values for each threshold and the solution to the game.

Based on the principles of the last few sections, we can parameterize the game this way:

For any positive integer n, $0 < x_{n+1} < y_{n+!} < x_n < y_n < 1$

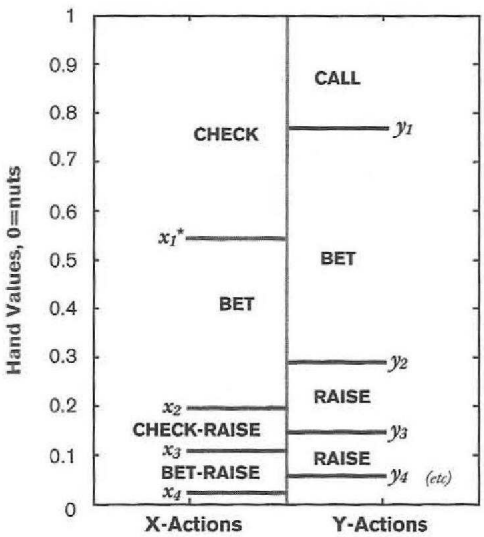

Figure 16.4. Strategy structure for [0,1] Game #8

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We can begin by writing indifference equations at the various *x* thresholds:

At $\boldsymbol{x}_1$:

$y_2 = 1 - y_1$ (This is unchanged from Game #6)

For X at x_2 (indifference between check-raising and betting):

Y's hand	X check-raises	X bets	Difference	Product
$[0, y_3]$	-3	-2	-1	$-y_3$
$[y_2, y_1]$	+2	+1	+1	$y_1 - y_2$
$[y_1, 1]$	0	+1	-1	$-(1 - y_1)$
Total				$\boldsymbol{-y_3 + 1 - 2y_1 - y_2}$

$-y_3 + 1 - 2y_1 - y_2 = 0$
$y_3 + y_2 + 1 = 2y_1$

Substituting:

$y_3 + 1 - y_1 + 1 = 2y_1$
$y_3 + 2 = 2y_1$

For X at x_3 (indifference between bet-re-raising and check-raising):

Y's hand	X bet-reraises	X check-raises	Difference	Product
$[0, y_4]$	-4	-3	-1	$-y_4$
$[y_3, y_2]$	+3	+2	+1	$y_2 - y_3$
$[y_2, y_1]$	+1	+2	-1	$-1(y_1 - y_2)$
$[y_1, 1]$	+1	0	+1	$1 - y_1$
Total				$\boldsymbol{-y_4 - y_3 + 2y_2 - 2y_1 + 1}$

$-y_4 - y_3 + 2y_2 - 2y_1 + 1 = 0$
$y_4 + (y_1 - y_2) = (y_2 - y_3) + (1 - y_1)$

For X at x_4 (indifference between check-raise-re-raising and bet-re-raising):

Y's hand	X bet-reraises	X check-raises	Difference	Product
$[0, y_5]$	-5	-4	-1	$-y_5$
$[y_4, y_3]$	+4	+3	+1	$y_3 - y_4$
$[y_3, y_2]$	+2	+3	-1	$-(y_2 - y_3)$
$[y_2, y_1]$	+2	+1	+1	$y_1 - y_2$
$[y_1, 1]$	0	+1	-1	$-(1 - y_1)$
Total				$\boldsymbol{-y_5 + 2y_3 - y_4 - 2y_2 + 2y_1 + 1}$

$-y_5 + 2y_3 - y_4 - 2y_2 + 2y_1 + 1 = 0$
$(y_3 - y_4) + ((y_1 - y_2) = y_5 + (y_2 - y_3) + (1 - y_1)$

More generally, at $\boldsymbol{x}_n$ we have the Following indifference equation when *n* is odd:

$y_{n+1} + (y_{n-2} - y_{n-1}) + (y_{n-4} - y_{n-3})\,[\ldots] + [y_1 - y_2] = (y_{n-1} - y_n) + (y_{n-3} - y_{n-2}) + [\ldots] + (1 - y_1)$

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And when *n is* even:

$(y_{n-1} - y_n) + (y_{n-3} - y_{n-2})$ […] + $[y_1 - y_2] = y_{n+1} + (y_{n-2} - y_{n-1}) + (y_{n-4} - y_{n-3})$ + […] + $(1 - y_1)$

Let's look at two actual thresholds, x_6 and x_7.
Writing the indifference equations for these yields:

$(y_5 - y_6) + (y_3 - y_4) + (y_1 - y_2) = y_7 + (y_4 - y_5) + (y_2 - y_3) + (1 - y_1)$

$y_8 + (y_5 - y_6) + (y_3 - y_4) + (y_1 - y_2) = (y_6 - y_7) + (y_4 - y_5) + (y_2 - y_3) + (1 - y_1)$

Subtracting the second of these equations from the first:

$y_8 = -2y_7 + y_6$

It's not too difficult to see that we can do the same thing for any value of *n*, such as k + 1 and k:

$y_{k+2} + (y_{k-1} - y_k) + (yk_{-3} - y_{k-2})$ […]+ $(y_1 - y_2) = (y_k - y_{k+1}) + (yk_{-2} - y_{k-1}) + (yk_{-4} - y_{k-3})$ […]+ $(1 - y_1)$

$(y_{k-1} - y_k) + (yk_{-3} - y_{k-2})$+ […]+ $(y_1 - y_2) = y_{k+1} + (y_{k-2} - y_{k-1}) + (yk_{-4} - y_{k-3})$+ […]+ $(1 - y_1)$

Again subtracting:

$y_{k+2} = -2y_{k+1} + y_k$

$y_{k+2} + 2y_{k+1} - y_k = 0$ **(16.1)**

Equations of this type are called ***difference equations*** and solving them requires a slightly more advanced mathematical technique. We present the derivation of the solution to Equation 16.1 in the Appendix to this chapter. The solution is that:

$y_n = r^n/(1 - r)$ (for y > 0)

This is Y's entire strategy, then, captured in a single equation, y_1 is 1/(1 - *r)*, and Y puts in the *(n* + l)th bet with a fraction of the hands with which he put in the nth bet equal to r. Compare this with Y's strategy in the game without check-raise. There, the factors of r separated each alternating threshold, from X to Y. So y_n was rx_{n-1}, and so on. Here, instead, the factors of *r* separate each of Y's thresholds from the next Y thresholds, so we have:

$y_n = ry_{n-1}$

With Y's strategy solved, we move on to X's strategy, which is solved in a similar manner but contains a little more complexity. Just as Y's strategy involved a pattern for specifying yn, we can specify xn. However, Y's strategy was unaffected by regions in thresholds with *n* much lower than the current one. X, on the other hand, alternates his regions between check-raising and betting initially, and this is the case for thresholds as low as y_1.

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For Y at y_1:

X's hand	Y check	Y bets
$[y_1, 1]$	0	+1
$[x_1, y_1]$	0	-1
$[x_3, x_2]$	0	-2
$[x_5, x_4]$	0	-2
$[x_7, x_6]$	0	-2
[…]		
$[x_{2k+1}, x_{2k}]$	0	-2

$$1 - y_1 = (y_1 - x_1) + 2(x_2 - x_3) + 2(x_4 - x_5) + 2(x_6 - x_7) + [...] + 2(x_{2k} - x_{2k+1}) \qquad (k \to \infty)$$

For this threshold, when the decision is between betting and checking, Y gains a bet when X has checked out of weakness, but loses two bets whenever X is planning to check-raise.

For Y at y_2:

X's hand	Y raises	Y calls
$[y_2, x_1]$	+2	+1
$[x_2, y_2]$	-2	-1
$[x_4, x_3]$	0	-2
$[x_6, x_5]$	0	-2
$[x_8, x_7]$	0	-2
[…]		
$[x_{2n}, x_{2n+1}]$	0	-2

$$x_1 - y_2 = (y_2 - x_2) + 2(x_3 - x_4) + 2(x_5 - x_6) + 2(x_7 - x_8) + [...] + 2(x_{2k} - x_{2k+1}) \qquad (k \to \infty)$$

And more generally, for Y at y_n:

X's hand	Y raises	Y calls
$[y_n, x_{n-1}]$	+2	+1
$[x_n, y_n]$	-2	-1
$[x_{n+2}, x_{n+1}]$	0	-2
$[x_{n+4}, x_{n+3}]$	0	-2
[…]		
$[x_{2n}, x_{2n+1}]$	0	-2

$$x_{n-1} - y_n = (y_n - x_n) + 2(x_{n+1} - x_{n+2}) + 2(x_{n+3} - x_{n+4}) + [...] + 2(x_{2k} - x_{2k+1}) \qquad (k \to \infty)$$

Returning briefly to y_1 we see that this equation is the general form of yn for all thresholds, if we define $x_0 = 1$ (which makes intuitive sense, as X will never bluff in a game where Y cannot fold.)

Consider this equation for a particular k and also for k + 1:

$$x_{k-1} - y_k = (y_k - x_k) + 2(x_{k+1} - x_{k+2}) + 2(x_{k+3} - x_{k+4}) + [...]$$
$$x_k - y_{k+1} = (y_{k+1} - x_{k+1}) + 2(x_{k+2} - x_{k+3}) + 2(x_{k+4} - x_{k+5}) + [\ldots]$$

Adding these two equations to each other, we find that the higher k terms cancel out neatly because of the alternating signs:

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$x_{k+1} - y_k + x_k - y_{k+1} = y_k - x_k + y_{k+1} + x_{k+}1$
$x_{k-1} + 2x_k - x_{k+1} = 2y_k + 2y_{k+1}$
$x_k + 2x_{k+1} - x_{k+2} = 2y_{k+1} + 2y_{k+2}$

Substituting for yn:

$x_k + 2x_{k+1} - x_{k+2} = 2(r^{k+1})/(l - r) + 2(r^{k+2})/(l - r)$
$r/(l - r) = (\sqrt{2} - l)/(2 - \sqrt{2}) = 1/\sqrt{2}$
$x_k + 2x_{k+1} - x_{k+2} = \sqrt{2}(r^k) + \sqrt{2}\,(r^{k+1})$
$x_k + 2x_{k+1} - x_{k+2} = \sqrt{2}\,(r^k)(l + r)$

$x_k + 2x_k + 1 - x_{k+2} = 2r^k$ **(16.2)**

This equation, too, can be solved using a more advanced mathematical technique, and the steps of the derivation are presented in the Appendix to this chapter. The solution to this equation yields X's strategy:

$x_n = r^{n-1}[(2r - l)(-l)^n + l]/2$

And so we have a recursive relationship that completely describes X's strategy.

Solution:

$x_n = r^{n-1}[(2r - l)(-l)^n + l]/2$

$y_n = r^n/(1 – r)$

The contrast between these two strategies is quite interesting: Y's strategy is relatively straightforward, with a simple exponential relationship between each threshold. Y puts in the first bet if checked to with hands below r/(l - r), and is willing to put m one additional bet with each multiple of r of his hands if X keeps raising.

Each X threshold lies between each of Y's thresholds, beginning with x_1 at 1 - r. However, these thresholds are not evenly spaced between Y's thresholds. X's check-raising regions are much narrower than his betting regions. This occurs in the formulas as the term in x_n with the $(-1)^n$ in it. The intuitive explanation for this is that when X has a very strong hand, he often wants to bet, because he gets a very large amount of value from the sequence "bet-call." As a result, X's thresholds oscillate in their relationship to Y's thresholds.

A partial table of the thresholds in the game

Threshold (n)	y_n	x_n	y_n/x_n
0	1	1	1
1	0.707107	0.585786	$r/(l – r) + {}^1/_2$
2	0.292893	0.171573	r/(l – r) + 1
3	0.12132	0.100505	$r/(l – r) + {}^1/_2$
4	0.050253	0.029437	r/(l – r) + 1
5	0.020815	0.017244	$r/(l – r) + {}^1/_2$

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$x_n = r^{n-1}[(2r - l)(-l)^n + l]/2$

The portion in brackets evaluates to 2r when n is even, and to 2(1 - *r)* when n is odd.

So $x_n = r^n$ when n is even and $r^{n-1}(1 - r)$ when n is odd.

We can find something interesting if we look at the size of x's checking regions:

$x_0 \rightarrow x_1 : 1 - (1 - r) = r$
$x_2 \rightarrow x_3 : r^2 - r^2(l - r) = r^2(l - (1 - r)) = r^3$
$x_4 \rightarrow x_5 : r^4 - r^4(1 - r) = r^4 (1 - (1 - r)) = r^5$
$x_{2n} \rightarrow x_{2n+l} : r^{2n} - r^{2n}(1 - r) = r^{2n+!}$

Summing the size of all these regions, we get the following expression

$$r \sum_{n=1}^{\infty} (r^2)^n$$

r^2 is less than 1, so the sum of this infinite series is r (1 / (1 - r^2)).

$$r^2 = 2 \cdot 2\sqrt{2} + 1 = 3 - 2\sqrt{2}$$
$$(\sqrt{2} - 1)/(1 - (3 - 2\sqrt{2})) = {}^1/_2$$

This intriguing result indicates that X checks half his hands overall. Most of these hands, of course, will reluctantly call (in fact, *r* of them). The remainder will check-raise and be ready-to put in additional bets as appropriate.

Key *Concepts*

- In full-street games with no folding, there is an important relationship between the hands with which the opponent could have taken an action, and the hands with which you will take your actions.
- If you cannot be raised and your opponent cannot fold, you should bet or raise with $^1/_2$ of the hands your opponent has bet or checked so far.
- If your opponent ***can*** raise you back, this ratio shrinks. In the case where an infinite number of raises are allowed, it becomes r.
- No-fold games are exercises in extracting the proper amount of value from strong hands.

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